

TRINETRA SECURITY INTELLIGENCE REPORT

System Documentation & Architectural Protocol

SYSTEM CLASSIFICATION: HIGH-ASSURANCE CORPORATE DEFENSE PROTOCOL

RELEASE CLASSIFICATION: PUBLIC / HACKATHON EVALUATION

DOCUMENT CONTROL NO: BOI-TRI-SEC-2026-V2.6

PROJECT CODENAME: TRINETRA (The Third Eye of Security)

1. Cover Page

Project Title

TRINETRA: Generative AI-Powered Fraudulent APK Detection, Risk Scoring & Banking Malware Investigation Platform

A Unified On-Device Static Decompilation, Bytecode Signature Search, and Explainable Forensic Intelligence Platform for Financial SecOps and Mobile Banking Defenses.

Team Specification: Team Pancha JANEYA

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Sponsoring Context

- **Hackathon:** PSB Cybersecurity, Fraud & AI Hackathon 2026
- **Host Institution:** Bank of India (BOI)
- **Design Core:** Enterprise Security Operations Center (SOC) Assistive Technology

2. Executive Summary

In the hyper-connected paradigm of digital retail banking, mobile application endpoints constitute the primary user boundary. Consequently, mobile devices have become the principal target for sophisticated, organized threat syndicates. Android-based banking Trojans have transitioned from simplistic credential harvesting overlays to highly autonomous, system-integrated malware families—such as TeaBot, Flubot, and SpyNote—that exploit native Android accessibility services to manipulate UPI transactions, intercept MFA tokens, and execute unauthorized fund exfiltrations.

TRINETRA is an industry-grade, local-first, on-device mobile forensic platform designed to bridge the structural capability gap in financial security operations centers (SOCs). Rather than routing suspicious binary files to remote, high-latency cloud sandboxes that delay incident response pipelines, TRINETRA achieves complete, localized static decompilation and DEX bytecode pattern signature scans within under 15 seconds. By leveraging decentralized intelligence rules linked to an advanced multi-layer risk scoring engine, TRINETRA matches technical bytecode artifacts to the standardized MITRE ATT&CK for Mobile matrix.

To assist frontline analysts, TRINETRA integrates an advanced Generative AI SecOps Co-Pilot powered by Google Gemini. This engine translates raw, obfuscated Smali class signatures and resource linkages into natural-language threat storylines, computes financial portfolio liabilities, and generates machine-executable isolation playbooks.

3. Problem Statement Analysis

3.1 Banking Fraud Challenges & Financial Loss Modeling

Retail banking systems are experiencing a surge in automated mobile-intercept frauds. The core tactical bottleneck is that current consumer protection structures are fundamentally reactive. TRINETRA addresses this lifecycle gap by identifying threats on the genuine consumer's registered device before unauthorized transfers occur.

3.2 Android Malware Threat Landscape

Modern banking malware utilizes several highly destructive attack mechanisms:

- **Automated Transfer Systems (ATS):** Leverages Accessibility API access to execute transfers directly inside legitimate banking applications.
- **Dynamic Overlays / Injectors:** Mimics login portals to capture credentials.
- **SMS Sniffing / MFA Bypass:** Intercepts incoming SMS payloads to extract OTP values.

3.3 The Need for On-Device Explainable AI Investigation

Traditional analysis tools fail to support analysts due to the "black box" problem, latency bottlenecks in web-based sandboxes, and a lack of actionable remediation playbooks.

4. Objectives

The strategic objectives outline the capabilities of the TRINETRA system:

- **High-Speed Static Deconstruction:** Analyze compiled Android (.apk) packages in under 15 seconds on a standard SOC mobile client.
- **Granular Bytecode Verification:** Inspect raw class implementations inside the `classes.dex` file structure.
- **Heuristics Rules Accuracy:** Maintain a high detection rate for complex banking Trojan indicators.
- **Explainable AI Integration:** Use advanced LLMs to translate raw bytecode structures into actionable incident reports.
- **Standardized MITRE Mapping:** Classify detected threat profiles against the standard **MITRE ATT&CK for Mobile v5** matrix.
- **Corporate Threat Isolation:** Generate actionable mitigation guidelines and asset-protection playbooks.

5. System Overview

TRINETRA operates on an **offline-first local architecture**. It runs entirely within a secure sandbox container on the operator's Android terminal, preventing data leakage during analysis.

System Architecture Layers

1. **Input Layer:** Collect APK files and initiate analysis via a physical sample upload or scenario simulator.
2. **Execution Layer:** Analyze the APK to identify malicious behavior through reverse engineering, static analysis, and AI-based classification.
3. **Output Layer:** Present actionable cybersecurity intelligence, including risk scores, MITRE mappings, and remediation recommendations.

6. Technical Deep Dive: Android APK Internals

An Android application package (APK) is a specialized, compressed ZIP archive. TRINETRA inspects key internal components:

Component	Format	Purpose
AndroidManifest.xml	Binary AXML	Defines permissions, services, and receivers.
classes.dex	Compiled Dalvik Bytecode	Scanned for malicious method hooks and API calls.
resources.arsc	Resource Tables	Non-compiled application layouts and strings.
META-INF / CERT.RSA	Security Keys	Contains public keys and signer certificates.

7. Malware Behavior & Risk Taxonomy

7.1 Banking Malware Behavior Matrix

Malware Family	Primary Delivery Vector	Key Permissions	Primary Fraud Objective
Anubis Banker	Malicious Droppers	RECEIVE_SMS	OTP Exfiltration & Overlay Attacks
TeaBot / Anatsa	Impostor Utility Apps	BIND_ACCESSIBILITY	Screen Scraping & Keylogging
FluBot	Smishing Links	READ_SMS, SEND_SMS	Mass SMS Exfiltration & Worm Spreading

Malware Family	Primary Delivery Vector	Key Permissions	Primary Fraud Objective
SpyNote Rat	Modified Droppers	RECORD_AUDIO, CAMERA	Remote Access & Audio Stealing

7.2 Android Permission Risk Taxonomy

Dangerous Permission	Risk Category	Potential Financial Impact
READ_SMS	Critical ▾	High risk of transaction hijacking via OTP theft.
BIND_ACCESSIBILITY	Critical ▾	Enables keylogging and automated background fraud.
SYSTEM_ALERT_WINDOW	Critical ▾	Enables credential theft via phishing overlays.
CAMERA / RECORD_AUDIO	High ▾	Risk of capturing secure QR codes or eavesdropping.

8. MITRE ATT&CK Mapping Framework

Action Identifier	MITRE ID	Classification	Technical Indicators
In-App Overlays	T1411	Credential Access	SYSTEM_ALERT_WINDOW; WindowManager.addView.
Accessibility Abuse	T1444	Privilege Escalation	BIND_ACCESSIBILITY_SERVICE registration.

Action Identifier	MITRE ID	Classification	Technical Indicators
Dynamic API Loading	T1425	Defensive Evasion	Imports of <code>DexClassLoader</code> or reflection.
SMS Exfiltration	T1437	Collection	<code>RECEIVE_SMS</code> receiver; <code>SmsManager</code> calls.

9. Results & Comparative Analysis

TRINETRA demonstrates significant operational efficiency over traditional frameworks:

Capability Checklist	TRINETRA	MobSF	Traditional Scanners
Analysis Platform	Mobile Android Tool	Docker Server App	Static Local Li... ▾
Triage Latency	Under 15 Seconds	5 to 10 Minutes	Instant (Signatu... ▾
MITRE Mapping	Full Mobile Matrix	Basic Text Lists	None ▾
Generative AI Analyst	Gemini Native	None	None ▾
Network Autonomy	Fully Offline-First	Local Host Only	Cloud Verification ▾

10. Conclusion

As digital banking adoption expands, securing mobile channels requires proactive, explainable security models. **TRINETRA** provides corporate security units with a mobile-first, high-performance static analysis platform. By combining speed, bytecode inspection, and generative analysis, TRINETRA enables cyber units to neutralize emerging threat vectors before they can impact financial systems, ensuring security and user trust.